

Chapter 0: A Brief Introduction to This Course

- Brief Introduction & Detailed Course Policy
 - [Lecture1: [slides](#)]

Chapter 1: Introduction

1. Basic Data Mining Concepts
 - [Lecture2: [slides_wo_Q](#)]
2. Getting to Know Big Data
 - [Lecture3: [slides_wo_Q](#)]
 - [Lecture4: [slides_wo_Q](#)]
 - [Lecture5: [slides_wo_Q](#)]
 - [Lecture6: [slides_wo_Q](#)]
3. Data Preprocessing
 - [Lecture7: [slides_wo_Q](#)]
 - [Lecture8: [slides_wo_Q](#)]
 - [Lecture9: [slides_wo_Q](#)]

Chapter 2: Finding Useful Patterns

1. Basic Concepts of Frequent Pattern
 - [Lecture10: [slides_wo_Q](#)]
2. The Apriori Algorithm
 - [Lecture11: [slides_wo_Q](#)]
3. The FP-growth Algorithm
 - [Lecture12: [slides_wo_Q](#)]
4. Concise Representations of Patterns
 - [Lecture13: [slides_wo_Q](#)]
5. Constraint-based Pattern Mining
 - [Lecture14: [slides_wo_Q](#)]
6. Significance of Patterns
 - [Lecture15: [slides_wo_Q](#)]

Mid-term Exam:

- **Time:** 3:30pm - 4:20pm, Feb. 27, 2026.
- **Contents covered:** all contents taught before the mid-term exam.
- **Format:** in-person in **our lecture room** during the lecture time for students not registered with SAS.
- **SAS accommodation:** SAS accommodated students please register the in-person exam with SAS.
- **More information:**
 - The mid-term exam is closed book.
 - This exam has two parts. Part 1 contains 10 multiple-choices selection questions. Part 2 contains 1 problem with 4 sub-questions to answer.
 - The format of part 2 is similar to your assignments.

Chapter 3: Clustering Analysis

- Basic Concepts and Partitioning Methods
 - [Lecture 16: [slides_wo_Q](#)]
- Hierarchical Clustering Methods
 - [Lecture 17: [slides_wo_Q](#)]
- Density-based Clustering
 - [Lecture 18: [slides_wo_Q](#)]
- Bi-clustering and Pattern-based Clustering
 - [Lecture 19: [slides_wo_Q](#)]
- Fuzzy Clustering, Mixture Models and EM Methods
 - [Lecture 20: [slides_wo_Q](#)]
- Dimensionality Reduction and Spectral Methods
 - [Lecture 21: [slides_wo_Q](#)]
- Clustering Evaluation
 - [Lecture 22: [slides_wo_Q](#)]

Chapter 4: Classification

1. Basic Concepts of Classification
 - [Lecture 23: [slides_wo_Q](#)]
2. Decision Trees and Overfitting
 - [Lecture 24: [slides_wo_Q](#)]
3. Classification Performance Evaluation
 - [Lecture 25: [slides_wo_Q](#)]
4. Bayesian Methods and Instance-based Methods
 - [Lecture 26: [slides_wo_Q](#)]
5. Linear Classification Methods
 - [Lecture 27: [slides_wo_Q](#)]
6. Support Vector Machines
 - [Lecture 28: [slides_wo_Q](#)]
7. Ensemble Methods
 - [Lecture 29: [slides_wo_Q](#)]
 - [Lecture 30: [slides_wo_Q](#)]
8. Neural Networks
 - [Lecture 31: [slides_wo_Q](#)]

Chapter 5: Business Intelligence

1. Data Warehousing

- [Lecture32: [slides_wo_Q](#)]
- [Lecture33: [slides_wo_Q](#)]

2. Data Lakes

- [Lecture34: [slides_wo_Q](#)]

3. Q&A Lecture of the course.

- [Lecture35: for question answering in the lecture room.]

Final Exam:

- **Time:** pending
- **Contents covered:** all contents taught.
- **Format:** in-person, Registrar's Office Supervised Exam.
- **SAS accommodation:** SAS accommodated students please contact SAS about the location and time of the exam.
- **More information:**
 - The final exam is closed book.
 - Please bring your calculator.
 - This exam has two parts. Part 1 contains 15 multiple-choices selection questions. Part 2 contains 4 problems, each of which includes multiple subquestions.
 - The format of the final exam is similar to your mid-term exam.